Cover Page

Name: Ava Shi

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Assignment: HW4

Due Date: 3/3/26

Acknowledgement: No Help

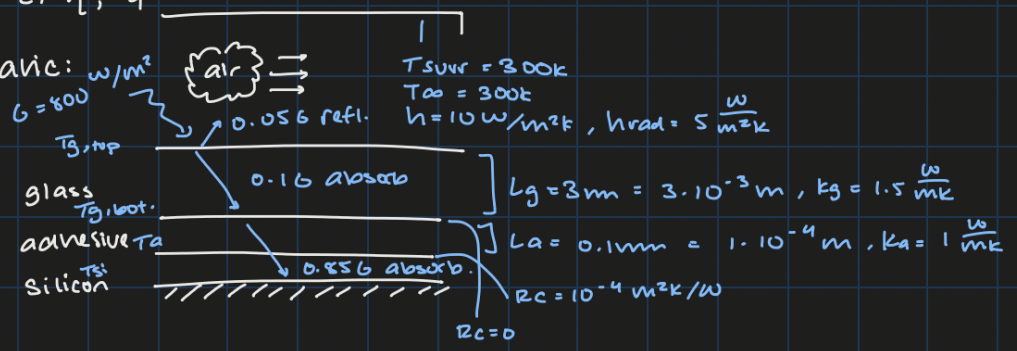
1) known: L_g, k_g, L_a, k_a, $\eta = a - bT_{si}$, R_c, G, $T_\infty = T_{sur}$, h, h_{rad}

find: a) res. net. + assumptions + calc. R_s

b) $T_{g,top}$, T_{si}

c) η, q''

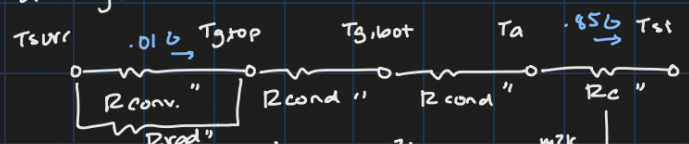
schematic:



assumptions: 1D, S.S

analysis:

a)



$\frac{1}{h} = .1 \frac{m^2K}{W}$ $\quad \frac{L_g}{k_g} = .002 \frac{m^2K}{W}$ $\quad \frac{L_a}{k_a} = .0001 \frac{m^2K}{W}$ given

$\frac{1}{h_{rad}} = .2 \frac{m^2K}{W}$

b) $q'' = h(T_{g,top} - T_{sur}) + h_{rad}(T_{g,top} - T_\infty)$, $q'' = 0.1G + (1-\eta)\,0.85G$

$\eta = 0.553 - .001T_{si}$

$T_{g,top} = 300 + \frac{q''}{h + h_{rad}} = 341.1K$

$= 536.7$

$T_{si} = T_{g,top} + q''R'' = 342.3K$

c) $\eta = 0.553 - 0.001T_{si} = 0.2107$, $q'' = \eta(.85G) = 143 \frac{W}{m^2}$

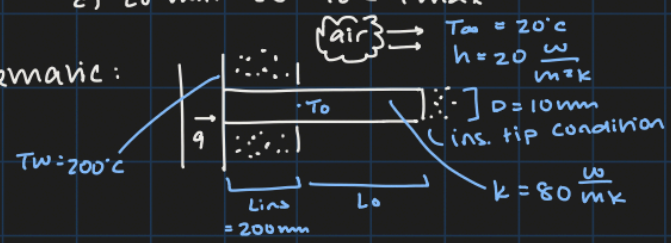
2) known: $D, k, T_w, L_{ins}, L_o, T_{max}, T_\infty, h$

find: a) T_o as fnc. geometry

b) $L_o = 200\,mm$, $T_o = ?$, $< T_{max}$?

c) L_o min st $T_o < T_{max}$

schematic:



$T_{max} = 100°C$

assumptions: 1D., s.s., ↓

analysis: a) adiabatic tip

$$q_{cond} = \frac{kA_c}{L_{ins}}(T_w - T_o) = q_{f,right} = \underbrace{M}_{\sqrt{hPkA_c}(T_o - T_\infty)}\tanh \underbrace{mL_o}_{\sqrt{\frac{hP}{kA_c}}}$$

$$A_c = \frac{\pi D^2}{4},\ P = \pi D$$

$$T_o = \frac{\frac{kA_c}{L_{ins}}T_w + \sqrt{hPkA_c}\tanh\left(\sqrt{\frac{hP}{kA_c}}L_o\right)T_\infty}{\frac{kA_c}{L_{ins}} + \sqrt{hPA_ck}\tanh\left(\sqrt{\frac{hP}{kA_c}}L_o\right)}$$

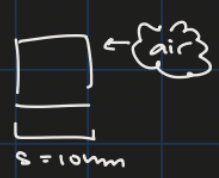
b) if $L_o = 200\,mm$, plug in $\rightarrow T_o = 81.47°C < T_{max}$

c) $100°C = \cdots$, graph $\rightarrow L_o = 73.3\,mm$

3) known: square, k, s, h, η_f

find: a) L, b) R_f & ε_f

schematic:



$h = 100\ \frac{W}{m^2K}$ $\quad k = 400\ \frac{W}{mK}$

$\eta_f = 0.6$

assumptions: 1D, s.s, convective tip, uniform A_c

analysis: $$\eta_f = \frac{q_f}{q_{max}} = \frac{M\left(\frac{\sinh mL + h/mk\cosh mL}{\cosh mL + h/mk\sinh mL}\right)}{hA_f\theta_b},\quad M = \sqrt{hs^2 4s \cdot 2}\,\theta_b,\quad m = \sqrt{\frac{h4s}{ks^2}}$$

solves w/ plug in #s $\rightarrow L = 149\,mm$

$$R_f = \frac{\theta_b}{q_f} = \frac{\cancel{\theta_b}}{\eta_f hA_f\cancel{\theta_b}} = \frac{1}{.6 \cdot 100 \cdot 4sL} = 2.76\ \frac{K}{W}$$

$$\varepsilon_f = \frac{q_f}{hA_b\theta_b} = \eta_f\frac{A_f}{A_c} = 34.3$$